

The Corporate Heist — Solution Documentation

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1. Summary

We treat the Vault as a different population from the Archive, not as more of the same. Three layers: (a) a forensic pass that removes internally inconsistent ("fabricated") profiles and collapses duplicate people; (b) a LightGBM merit model trained on post_hire_score with every feature the debrief says the new panel ignores deliberately removed; (c) a regime-shift layer that turns each line of the debrief into a hard rule or a calibrated bonus. Validated end-to-end on the Ledger: the de-biased ("vault-lens") model recovers 68 of that cycle's 150 true top-5% at k=150 (chance = 7.5); the full model with the deprecated old-panel features recovers 95 — which is exactly the +27 inflation we expect from biases the new panel no longer uses, and the reason we do not submit the full model's ranking.

2. Data cleaning and parsing

Every numeric column is multi-format. technical_assessment appears as 47, 49/100, 0.45, 43.0 %, absent, not taken — we map the fraction form (≤ 1.0) by $\times 100$ and keep a separate "did not sit the test" flag rather than imputing a zero. aptitude_score appears as 6.3, 5.0/10, 41%. last_rating mixes 3, 3/5, 3.0 and the words Outstanding / Exceeds / Meets / Needs Improvement / Below Expectations, plus New joiner - not rated which is missingness, not a 0. total_experience appears as 9.4 years, 9.4 yrs, >20, 20+ years, 135 months, Fresher, <1 year. Compensation is the worst: 13.9 LPA, 13.9 lpa, Rs13,90,000, 1390000, 13.90L, INR 13.9 lakh, 7.53 Cr and ** \$22,369 ** — we normalise everything to INR lakh p.a. and convert USD at 83 INR/USD. career_path uses three separators (> , -> , | , plus a unicode arrow) and two duration syntaxes ([23 mo] , (1.2 yrs)). institute collapses 730 raw spellings to 314 schools (~~III Delhi~~ $\rightarrow$ ~~I.I.T. Delhi~~ $\rightarrow$ ~~III-D~~ $\rightarrow$ ~~IIIT~~ $\rightarrow$ ~~Indian Institute of Technology Delhi~~); current_city merges Madras $\rightarrow$ Chennai, Gurgaon $\rightarrow$ Gurugram. kpi_met has ten spellings of a boolean. public_code_contributions (test-only) appears as 12, 3, 10+, 3 PRs, 3 merged PRs, not tracked — we parse the first integer and leave not tracked/blank as missing, never as zero.

3. What drives a great hire — our findings

Spearman correlation with post_hire_score on the Archive: technical_assessment +0.27, last_rating +0.20, aptitude_score +0.20, graduation_year +0.13; total_experience -0.12 and current_ctc -0.12 are *negative*. kpi_met is the strongest binary: mean 53.4 vs 46.5, P(top 5%) 0.081 vs 0.033. Awards lift the mean to 53.8 (Spot Award 56.9). Seniority is inverted-U: candidates at individual-contributor levels average 51.3–51.4 while Lead/Principal/Director/VP applicants average 45.6–46.3. trainings_last_year (+0.003) and training_hours (+0.017) are noise. The recruiter_note field is 1,090 concatenations of ~25 fixed sentences; we count positive atoms ("owned production incidents", "shipped a feature used by 1M+ users") against negative ones ("reference check was lukewarm", "missed two sprint commitments") rather than embedding the text.

4. The current cycle

The debrief describes four changes and all four are visible in the data. (i) **Notice period.** notice_period > 60 days occurs **208 times in test.csv and zero times in train.csv or dev.csv** — 90/120/150/180-day and 3–6-month values exist only in the Vault. Excluded. (ii) **Title inflation.** We learn, per seniority level, the minimum tenure ever observed in the Archive (e.g. Director/VP levels never appear below ~13 years of combined experience-or-time-since-graduation). The Vault contains **149 profiles** that violate their level's learned floor. Excluded. (iii) **Public code contributions.** Test-only. Median 6, p90 26, max 408; uncorrelated with experience (-0.02) and with our fabrication flags, i.e. an independently injected signal. Following "a handful means nothing, dozens gets fast-tracked" we apply a

log-ramp bonus that is zero below 12, saturating at +1.0 z by 48. `not_tracked` receives neither bonus nor penalty.

(iv) **Deprecated preferences.** We drop `institute`, `current_city`, `recruitment_channel`, `company_type`, `company_size`, degree and the career-gap feature from the Vault model. This is costly and deliberate: on the Ledger those features inflate `recall@150` from 68 to 95. They are exactly the biases the new panel abandoned. The one survivor is the old-boys' network: regressing `post_hire_score` on `merit` + `role` and grouping the residual by school ($n \geq 40$, $t \geq 3$) isolates a handful — MDS Ajmer, IISc, BITS Pilani, IIT Kharagpur, NIT Warangal. Note that MDS Ajmer is not a conventional prestige school yet carries the largest residual, which is the signature of a leadership-alumni effect rather than brand recognition. Those five get a modest +0.25 z bonus — "a leg up", not a ticket. (v) **No pedigree, no problem.** A meaningful share of Vault rows come from institutes that appear nowhere in the Archive or Ledger, with a materially higher median technical assessment than the rest of the pool. New-school candidates in the top decile of their role's assessment, with a coherent career history, get a +0.70 z bonus.

5. Profiles we excluded and why

Fabricated (393 in test; 561 in train used only to clean the training label). A profile is rejected if any internal check fails: age at graduation < 18; experience > age - 20; experience > years since graduation + 1.5; stated experience exceeding the career history by > 1 year; career history overrunning stated experience by > 8 years or graduation by > 4 years; more than 10 unaccounted years. Calibration on the Archive: 561 rows flagged, mean `post_hire_score` **5.49**, and **not one of them is in the true top 5%**. On the Ledger, 87 rows flagged and **zero of the 150 winners** among them. The thresholds sit in empty bands — the Archive has no row with age-at-graduation between 17 and 19. **Cannot join in time (208).** `notice_period` > 60 days. **Title inflation (149).** As above. **Duplicate people (246 rows in test).** Union-find over normalised phone (last 10 digits), e-mail (lower-cased, dots stripped from the local part) and sorted-name + graduation year. 10,000 rows resolve to **9,754 people**; we keep each person's best-scoring entry. Example: CH-0R5RQA "Advani, Vaagdevi" / 61047 27758 and CH-CEZTFK "Vaagdevi Advani" / +91 6104727758 are one IIT Delhi 2013 DevOps applicant.

6. Model and selection procedure

LightGBM on the de-fabricated Archive (19,439 rows), ~40 engineered features, bagged over seeds 42/202/777. Two heads: an L2 regressor on `post_hire_score` and a binary classifier on "is in the top 5%". Their percentile ranks are averaged 50/50. The blended rank is z-scored, the Vault bonuses (\$4) are added, hard exclusions (\$5) are set to -infinity, duplicate people are collapsed to their best-scoring entry, and the top 500 are emitted in descending order. Validation is the Ledger: the same code path, the same forensics, `recall@150` printed at run time as a build-time assertion, not a claim.

7. What did not work

- **Skill-content features.** Lift of each skill token among top performers per role was 1.3–2.2 on thin counts and incoherent (`pytorch` ranked top for DevOps/SRE, `vuejs` second). Only the *count* of skills survived into the feature set.
- **Sentence-transformer embeddings of recruiter_note.** The field is a closed vocabulary of ~25 atoms; keyword counting is exact and costs nothing, embeddings cost real runtime budget for no measurable gain.
- **Target-encoding the institute for the Vault.** Lifts Ledger recall by ~27 points, but it is the deprecated bias in disguise. Rejected on the debrief's evidence, not on taste.
- **Treating not_tracked public contributions as 0.** Would penalise over a thousand candidates for a recruiter's data-entry habit rather than their actual record.
- **Predicting a score for all 10,000 and cutting at the 95th percentile.** Only membership and order of the 500 matter; effort was better spent on the exclusion rules, which is where the fabricated/duplicate/inflated profiles actually get removed.

8. External resources and AI tools (mandatory)

- **Libraries:** numpy, pandas, scikit-learn, LightGBM (all from the permitted list). No external datasets, no pretrained models, no network access.
- **Reference constants:** USD→INR at 83 (public reference rate) for \$ -denominated CTC; institute and city alias tables were written by hand from the values present in the provided files, not from any external list.
- **AI assistants:** Anthropic Claude (Claude Code) was used to draft and refactor the parsing code, to run the exploratory analysis behind §3–§5, and to draft this document. Every rule and threshold reported here was re-derived and verified by the team against train.csv and dev.csv; all numbers quoted are reproducible from the submitted code and are printed by the code itself at run time.

9. How to run

```
python main.py
```

train.csv , dev.csv , dev_winners.csv , test.csv in the working directory; writes submission.csv there. Python 3.11, CPU only. **Runtime ~60 s** (limit 5 min, i.e. 300 s). Seeds: SEED = 42 , model seeds 42/202/777, PYTHONHASHSEED=42 . Two consecutive runs produce byte-identical output (verified: identical MD5 hash).